

ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Assignment 4 - Due date 02/10/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github. And to do so you will need to fork our repository and link it to your RStudio.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., “LuanaLima_TSA_A04_Sp26.Rmd”). Then change “Student Name” on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: “xlsx” or “readxl”, “ggplot2”, “forecast”, “tseries”, and “Kendall”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
knitr::opts_chunk$set(tidy = TRUE, tidy.opts = list(width.cutoff = 20))
library(forecast)
library(Kendall)
library(readxl)
library(tidyverse)
library(tseries)
```

Questions

Consider the same data you used for A3 from the spreadsheet. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. **For this assignment you will work only with the column “Total Renewable Energy Production”.**

```
outlook <- read_excel("./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx",
  sheet = "Monthly Data",
  skip = 10)
outlook <- outlook[-c(1),
  ]
outlook <- outlook |>
  select("Month", "Total Renewable Energy Production")
outlook <- outlook |>
```

```
mutate(across(c("Total Renewable Energy Production"),
  as.numeric))
outlook <- outlook |>
  rename(original = `Total Renewable Energy Production`)
```

Stochastic Trend and Stationarity Tests

For this part you will work only with the column Total Renewable Energy Production.

Q1

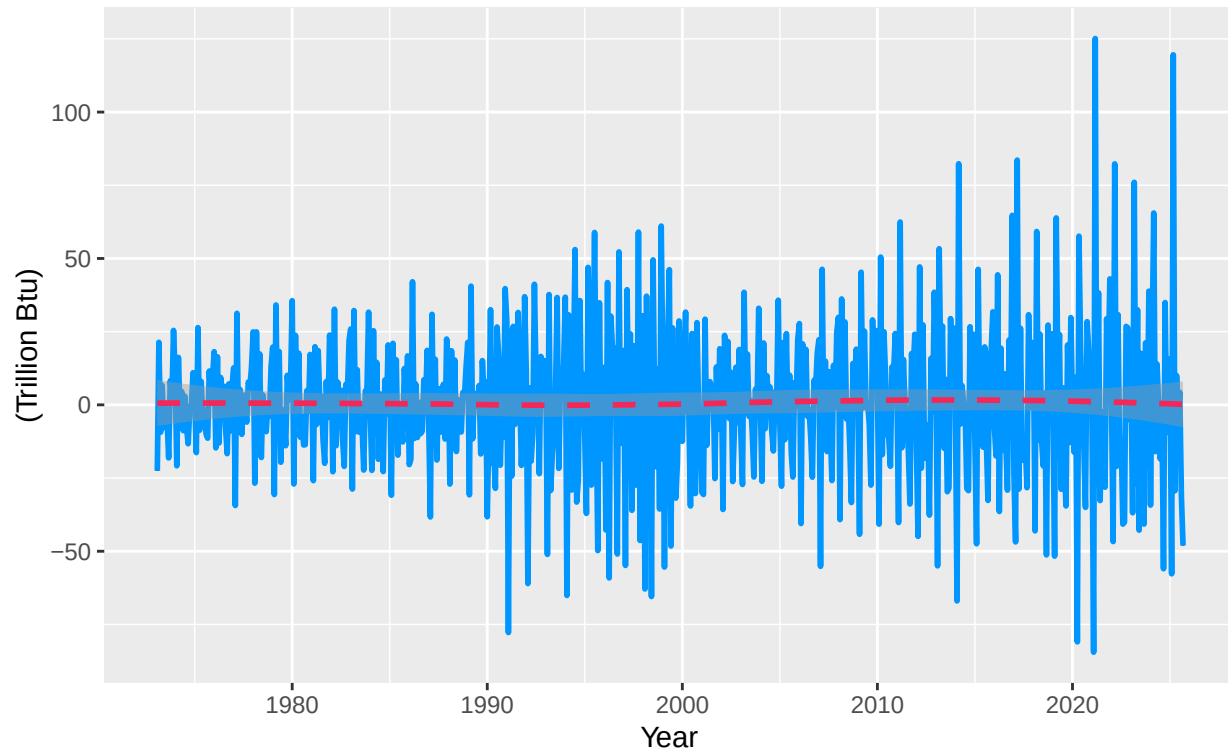
Difference the “Total Renewable Energy Production” series using function `diff()`. Function `diff()` is from package `base` and take three main arguments: * *x* vector containing values to be differenced; * *lag* integer indicating with lag to use; * *differences* integer indicating how many times series should be differenced.

Try differencing at lag 1 only once, i.e., make `lag=1` and `differences=1`. Plot the differenced series. Do the series still seem to have trend?

```
outlook <- outlook |>
  mutate(differenced = c(NA,
    diff(original,
      lag = 1,
      differences = 1)))
t <- c(1:633)
renewable_first_model <- lm(outlook$original ~
  t)
beta0_renewable_first <- as.numeric(renewable_first_model$coefficients[1])
beta1_renewable_first <- as.numeric(renewable_first_model$coefficients[2])
outlook <- outlook |>
  mutate(detrended = original -
    (beta0_renewable_first +
      beta1_renewable_first *
        t))
history <- ts(outlook[,
  -c(1)], start = c(1973,
  1), frequency = 12)
p1 <- autoplot(history[,
  2]) + geom_line(color = "#0096FF",
  linewidth = 1) +
  geom_smooth(color = "#F52757",
    linewidth = 1,
    linetype = "dashed") +
  theme_update() +
  labs(title = "Time Series For Total Renewable Energy Production",
    subtitle = "Differenced Data From 1973 To 2025",
    x = "Year", y = "(Trillion Btu)")
p1
```

Time Series For Total Renewable Energy Production

Differenced Data From 1973 To 2025



Answer: This series does not have an apparent trend.

Q2

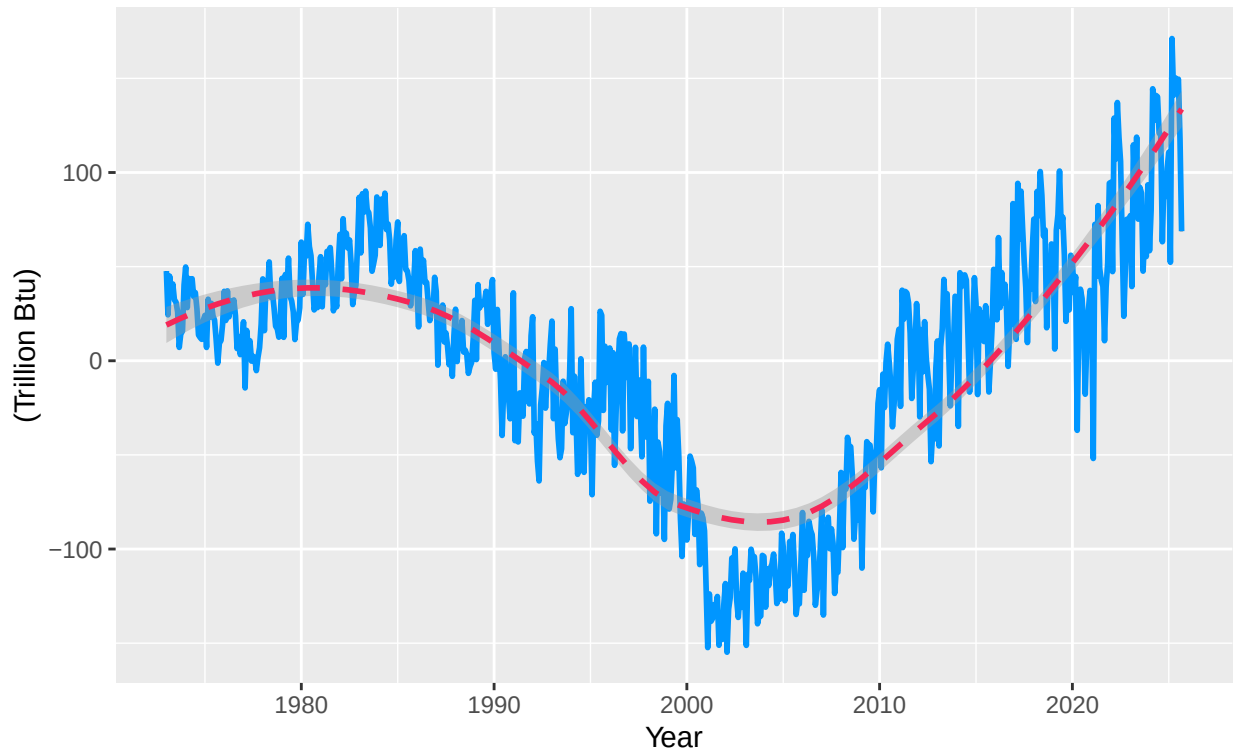
Copy and paste part of your code for A3 where you run the regression for Total Renewable Energy Production and subtract that from the original series. This should be the code for Q3 and Q4. make sure you use assign same name for the time series object that you had in A3, otherwise the code will not work.

```
p2 <- autoplot(history[,
  3]) + geom_line(color = "#0096FF",
  linewidth = 1) +
  geom_smooth(color = "#F52757",
  linewidth = 1,
  linetype = "dashed") +
  theme_update() +
  labs(title = "Time Series For Total Renewable Energy Production",
  subtitle = "Detrended Data From 1973 To 2025",
  x = "Year", y = "(Trillion Btu)")
```

p2

Time Series For Total Renewable Energy Production

Detrended Data From 1973 To 2025



Q3

Now let's compare the differenced series with the detrended series you calculated on A3. In other words, for the "Total Renewable Energy Production" compare the differenced series from Q1 with the series you detrended in Q2 using linear regression.

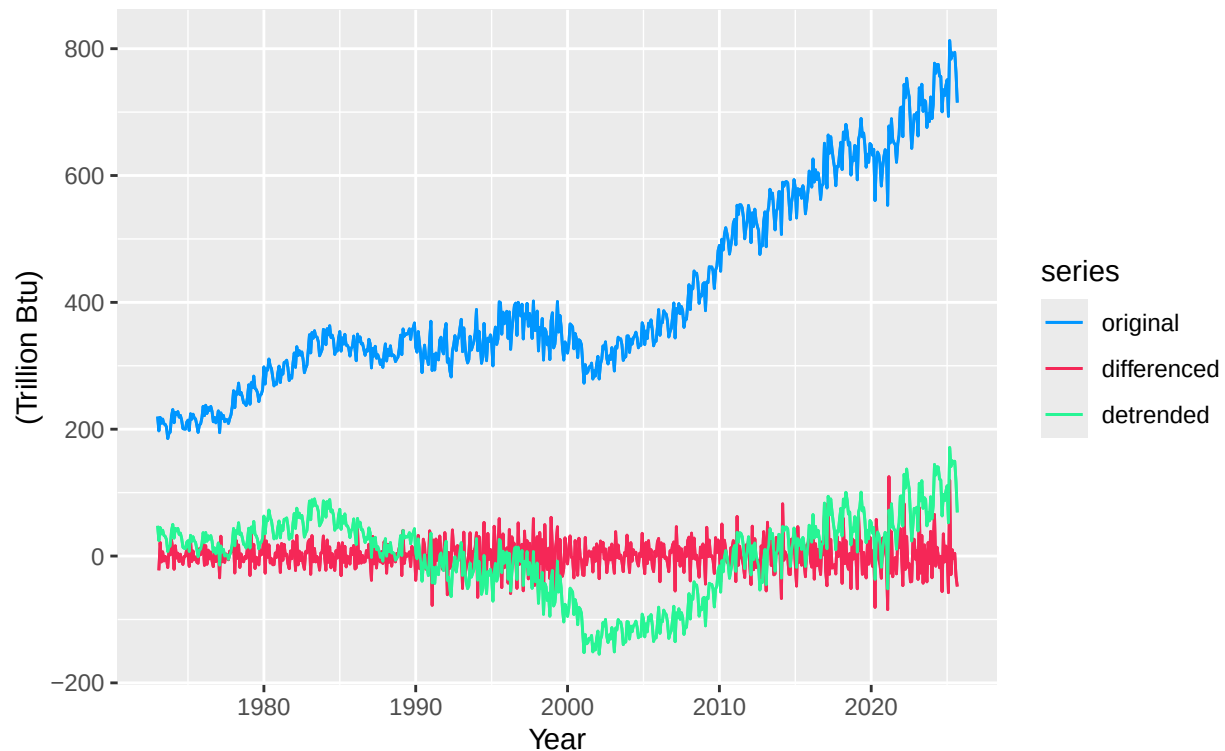
Using `autoplot()` + `autolayer()` create a plot that shows the three series together (i.e. "Original", "Differenced", "Detrended lm()"). Make sure your plot has a legend. The easiest way to do it is by adding the `series=` argument to each `autoplot` and `autolayer` function. Look at the key for A03 for an example on how to use `autoplot()` and `autolayer()`.

What can you tell from this plot? Which method seems to have been more efficient in removing the trend?

```
p3 <- autoplot(history) +
  geom_line(linewidth = 0.5) +
  scale_color_manual(values = c("#0096FF",
    "#F52757", "#27F595")) +
  theme_update() +
  labs(title = "Comprehensive Time Series For Total Renewable Energy Production",
    subtitle = "Data From 1973 To 2025",
    x = "Year", y = "(Trillion Btu)")
```

p3

Comprehensive Time Series For Total Renewable Energy Production Data From 1973 To 2025

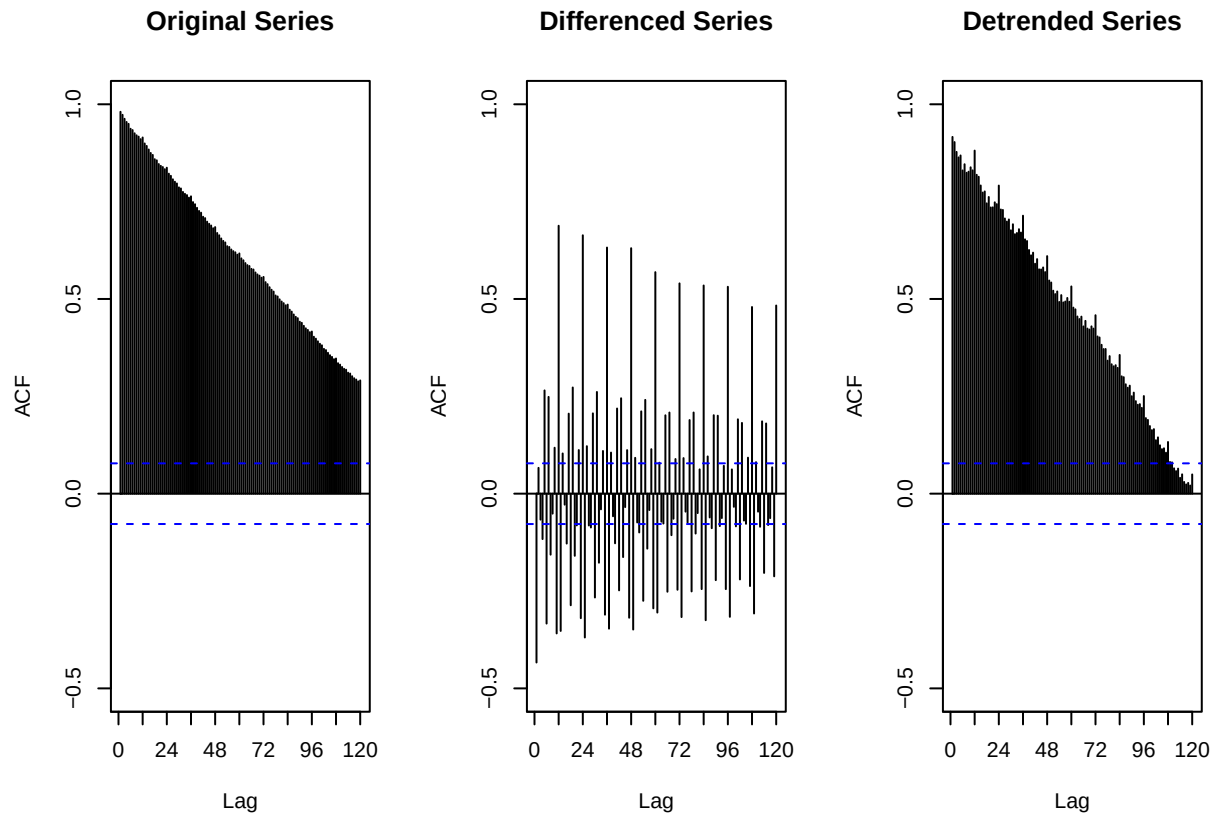


Answer: The original data indicates that total renewable energy production has an upward trend. The detrended data displays a quadratic trend. Thus, differencing is the more efficient method to remove trends.

Q4

Plot the ACF for the three series and compare the plots. Add the argument `ylim=c(-0.5,1)` to the `autoplot()` or `Acf()` function - whichever you are using to generate the plots - to make sure all three y axis have the same limits. Looking at the ACF which method do you think was more efficient in eliminating the trend? The linear regression or differencing?

```
par(mfrow = c(1, 3))
Acf(history[, 1], lag.max = 120,
     ylim = c(-0.5, 1),
     type = "correlation",
     plot = TRUE, main = paste("Original Series"))
Acf(history[, 2], lag.max = 120,
     ylim = c(-0.5, 1),
     type = "correlation",
     plot = TRUE, main = paste("Differenced Series"))
Acf(history[, 3], lag.max = 120,
     ylim = c(-0.5, 1),
     type = "correlation",
     plot = TRUE, main = paste("Detrended Series"))
```



Answer: Differencing might be the more efficient method to eliminate trends.

Q5

Compute the Seasonal Mann-Kendall and ADF Test for the original “Total Renewable Energy Production” series. Ask R to print the results. Interpret the results for both test. What is the conclusion from the Seasonal Mann Kendall test? What’s the conclusion for the ADF test? Do they match what you observed in Q3 plot? Recall that having a unit root means the series has a stochastic trend. And when a series has stochastic trend we need to use differencing to remove the trend.

```
MK_first <- SeasonalMannKendall(history[,
  1])
print("Results for Seasonal Mann Kendall Test (Monthly Data)")
```

```
## [1] "Results for Seasonal Mann Kendall Test (Monthly Data)"
```

```
print(summary(MK_first))
```

```
## Score = 13083 , Var(Score) = 201135
## denominator = 16379.5
## tau = 0.799, 2-sided pvalue =< 2.22e-16
## NULL
```

```
print("Results for ADF Test (Monthly Data)")
```

```
## [1] "Results for ADF Test (Monthly Data)"
```

```
print(adf.test(history[,  
  1], alternative = "stationary"))
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: history[, 1]  
## Dickey-Fuller = -1.0247, Lag order = 8, p-value = 0.9347  
## alternative hypothesis: stationary
```

Answer: The Seasonal Mann Kendall Test indicates that the original series of total renewable energy production follows a trend. The ADF Test indicates that the original series of total renewable energy production is stochastic. Therefore, differencing is the preferred method to eliminate trends.

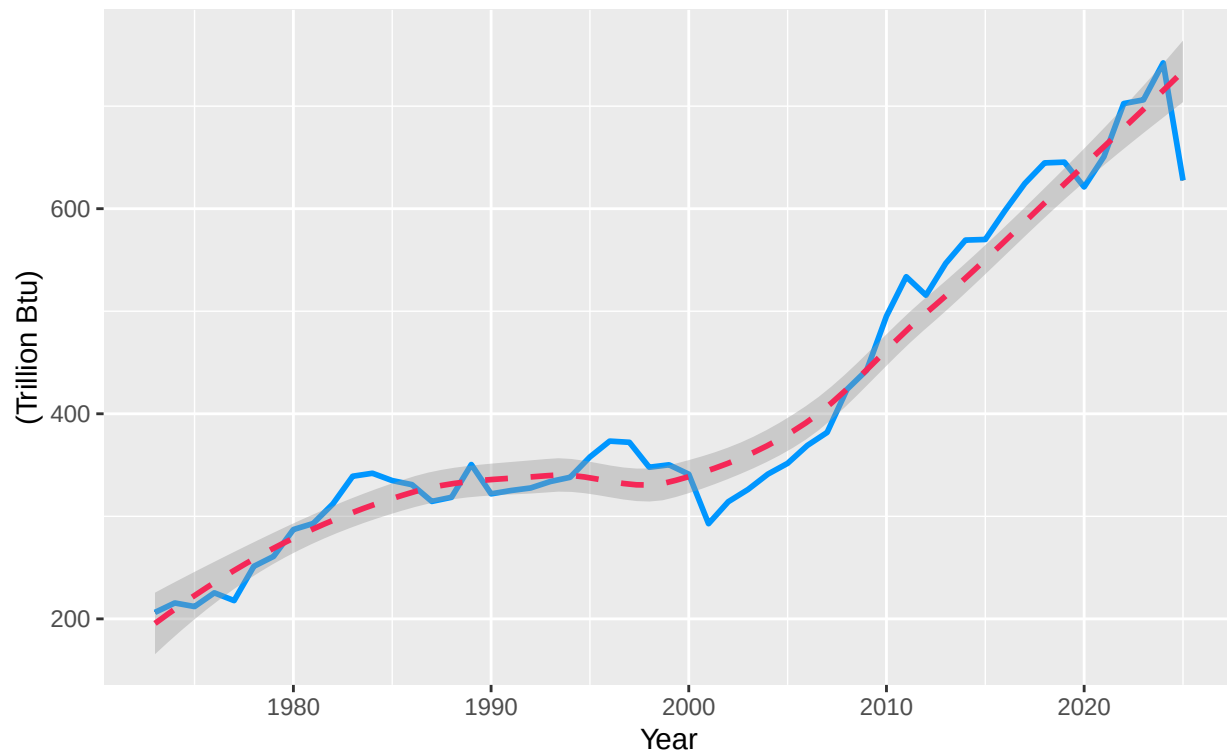
Q6

Aggregate the original “Total Renewable Energy Production” series by year. You can use the same procedure we used in class. Store series in a matrix where rows represent months and columns represent years. And then take the columns mean using function `colMeans()`. Recall the goal is to remove the seasonal variation from the series to check for trend. Convert the accumulated yearly series into a time series object and plot the series using `autoplot()`.

```
renewable_matrix <- matrix(history[,  
  1], byrow = FALSE,  
  nrow = 12)  
neat <- colMeans(renewable_matrix)  
duration <- c(year(first(outlook$Month)):year(last(outlook$Month)))  
aggregate_outlook <- data.frame(duration,  
  neat)  
aggregate_history <- ts(aggregate_outlook[,  
  -c(1)], start = 1973,  
  frequency = 1)  
p4 <- autoplot(aggregate_history) +  
  geom_line(color = "#0096FF",  
    linewidth = 1) +  
  geom_smooth(color = "#F52757",  
    linewidth = 1,  
    linetype = "dashed") +  
  theme_update() +  
  labs(title = "Time Series For Total Renewable Energy Production",  
    subtitle = "Annual Average Data From 1973 To 2025",  
    x = "Year", y = "(Trillion Btu)")
```

p4

Time Series For Total Renewable Energy Production
Annual Average Data From 1973 To 2025



Q7

Apply the Mann Kendall, Spearman correlation rank test and ADF. Are the results from the test in agreement with the test results for the monthly series, i.e., results for Q5?

```
MK_second <- MannKendall(aggregate_history)
print("Results for Mann Kendall Test (Yearly Data)")
```

```
## [1] "Results for Mann Kendall Test (Yearly Data)"
```

```
print(summary(MK_second))
```

```
## Score = 1124 , Var(Score) = 16995.33
## denominator = 1378
## tau = 0.816, 2-sided pvalue =< 2.22e-16
## NULL
```

```
print("Results for Spearman Correlation Test (Yearly Data)")
```

```
## [1] "Results for Spearman Correlation Test (Yearly Data)"
```



```
SP <- cor.test(neat,
  duration, method = "spearman")
print(SP)
```

```
##
## Spearman's rank correlation rho
##
## data: neat and duration
## S = 1898, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.9234801
```

```
print("Results for ADF Test (Yearly Data)")
```

```
## [1] "Results for ADF Test (Yearly Data)"
```

```
print(adf.test(aggregate_history,
  alternative = "stationary"))
```

```
##
## Augmented Dickey-Fuller Test
##
## data: aggregate_history
## Dickey-Fuller = -1.6789, Lag order = 3, p-value = 0.7037
## alternative hypothesis: stationary
```

Answer: Tests on yearly data reflect the same information. The original series of total renewable energy production has a strong trend.